Mohammad Naqvi

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Experience

InfluxData March 2024 – April 2026
Software Engineer *Remote, MI*

- Drove architecture decisions and built InfluxDB's licensing enforcement system end-to-end for air-gapped deployments, including a Rust-based server and Kubernetes controller for signed-JWT issuance and offline dual-layer verification, a Go CLI for support tooling, and a Grafana dashboard for on-call visibility.
- Investigated data-lifecycle internals across the catalog and storage layers to shape deletion policy, and shipped TypeScript/React Admin UI features while cutting redundant API calls by 40% through client-side caching.
- Folded token management and table schema retrieval into a single Rust monolith (dozens of REST endpoints, OpenAPI-specced) as part of a company-wide move off a sprawl of small Go services.
- Operated production Kubernetes delivery on AWS and GKE with GitHub Actions and Grafana.

Intuitive Aug 2023 – Dec 2023
Software Engineer Intern *Sunnyvale, CA*

- Decomposed a manufacturing floor's equipment state into a hierarchy of machines within a line and lines within a factory, in TypeScript/React, so a status change at one machine correctly propagated up to how its line and factory were reported.

Tesla May 2023 – Aug 2023
Software Engineer Intern *Austin, TX*

- Rewrote a Kafka-fed materials-sync service from C# to Go, moving its integration with Dassault's 3DEXPERIENCE platform off a fragile C++ daemon process communicating over a TCP socket and onto an in-process foreign function interface.
- Ported the material-classification domain model (metals, polymers, composites, each with distinct property sets), along with the validation rules that determine which incoming records are CAD-relevant before they sync.

Microsoft June 2022 – Sept 2022
Software Engineer Intern *Redmond, WA*

- Implemented an adapter layer in C# decoupling an API's public interface from its backend, letting a legacy service transparently migrate to a new Redis-backed data store without breaking existing callers.
- Built a cost model from production usage logs that scored each legacy API by migration readiness, projecting the adapter-based migration strategy at roughly a third of the effort of migrating each caller individually.

Education

University of Michigan, Ann Arbor Ann Arbor, MI
M.S. Computer Science and Engineering *April 2027*

- **Qualcomm Master's Fellowship** ($15,000) — Rackham Graduate School, 2026

San Jose State University San Jose, CA
B.S. Computer Science *December 2023*

Relevant Coursework: Computer Architecture, Advanced Databases, Parallel Computer Architecture, Formal Verification of Systems Software, Programmable Data Planes for In-Network ML

Projects

Directed Study — Database Systems | *Rust* | University of Michigan

- Independent study on database query optimization.

pq-vector | *Rust, Apache Arrow, DataFusion* | github.com/mohammadnaqvi04/pq-vector

- Extended a Parquet-based vector search library with temporal file pruning, embedding timestamp bounds in Parquet footer metadata to enable predicate pushdown that skips entire files before any IVF index access.

Technical Skills

Languages: Rust, Python, C++, Go, TypeScript, SQL, C#
Technologies: Axum, Tokio, gRPC, PostgreSQL, Docker, Kubernetes, AWS, GKE, Apache Kafka, JWT, React, CI/CD